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6. (a) What is significance of pentose phosphate pathway? (6)
- (b) Describe the process through which glucose is stored in the body. (6)
7. Write short notes on **any three** :- (4×3)
- (i) Malate aspartate shuttle
- (ii) Oxidative phosphorylation
- (iii) Intermediary metabolism
- (iv) ETC

(500)

[This question paper contains 4 printed pages.]

05.06.24(M)

Your Roll No.

Sr. No. of Question Paper : 3121

Unique Paper Code : 32231403

Name of the Paper : Biochemistry of Metabolic Processes

Name of the Course : B.Sc. (H) Zoology, LOCF

Semester : IV

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Attempt any **FIVE** QUESTIONS in all, question no. **1** is **COMPULSORY**.

1. Define the following : (5)

- (i) Cori cycle
- (ii) Fermentation
- (iii) Glycogenolysis
- (iv) Oxidative phosphorylation
- (v) Acidosis

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(b) Differentiate between the following : (2×5=10)

- (i) Catabolism and anabolism
- (ii) Inhibitors and uncouplers
- (iii) Transamination and deamination
- (iv) Hexokinase and glucokinase
- (v) Glucogenic and ketogenic amino acids

(c) Give the function(s) of the following enzymes :
(1×4=4)

- (i) Transaldolase
- (ii) Succinate dehydrogenase
- (iii) Methyl malonyl CoA mutase
- (iv) Phosphoenolpyruvate carboxykinase

(d) Expand the following : (1×5=5)

- (i) UDP Glucose
- (ii) PFK
- (iii) NADPH
- (iv) DHAP
- (v) GTP

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(e) Draw the structures of the following : (3)

- (i) α -Ketoglutarate
- (ii) Glutamate
- (iii) HMG CoA

2. (a) With the help of structural formulae, describe the fate of glucose under aerobic condition. (8)

(b) Add a note on ketone bodies. (4)

3. (a) "Gluconeogenesis is not the reversal of glycolysis", justify the statement. (6)

(b) Explain the chemiosmotic hypothesis. (6)

4. (a) Give a detailed account of urea cycle. Add a note on its link to TCA cycle. (8)

(b) Enlist the cascade of metabolic events occur during starving condition. (4)

5. Describe the β -oxidation of palmitic acid. Discuss about fates of the end products produced in this process. (12)

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